Name:…… Index No……..……

P525/1
CHEMISTRY
Paper 1
July/August 2019
2¾ hours

MWALIMU EXAMINATIONS BUREAU

UACE RESOURCE MOCK EXAMINATIONS 2019

CHEMISTRY

PAPER 1

2 Hours 45 Minutes

Instructions to Candidates

*Answer **all** questions in section **A** and **six** questions in section **B**.*

All questions must be answered in the spaces provided.

The Periodic Table, with relative atomic masses, is supplied.

Mathematical tables (3-figure table) are adequate or non-programmable scientific electronic calculators may be used.

Illustrate your answers with equations where applicable.

Where necessary, use the following:

Molar gas constant, R $= 8.31\ JK^{-1}mol^{-1}$.

Molar volume of gas at s.t.p is 22.4 litres.

Standard temperature $= 273\ K$

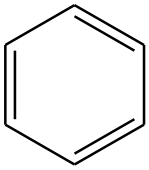
Standard pressure $= 101325\ Nm^{-2}$

	For Examiners' use Only															
1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17

Section A: (46 marks)

Answer **all** questions from this section

1. Complete the following reaction equations and write the IUPAC names of the main organic product(s) in each case.

(a) 

$\xrightarrow[\text{u.v light}]{Cl_2}$ .....(01 mark)

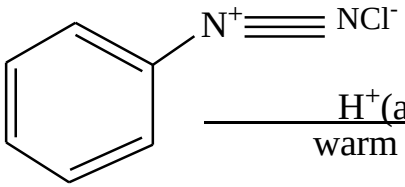
Name of product..... (0½ mark)

(b) $(CH_3CO_2)_2Ca \xrightarrow{heat}$ … … … … … … … … … … . … … … … … … … . … … …(01 mark)

Name of product.....(0½ mark)

(c) $C_6H_5COOCH_2CH_3 \xrightarrow[(ii)H_2O(l)]{(i)LiAlH_4/dry\ ether}$ … … … … … … … … … . . … ….(01 mark)

Name of product(s).....(1 mark)

(d) 

$\xrightarrow[\text{warm}]{H^+(aq)}$ .....(01 mark)

Name of product.....(0½ mark)

2. (a) State **Raoult's law**. (01 mark)

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(b) A liquid mixture contains 15.6 g of benzene and 55.2 g of methylbenzene at 25°C. The vapour pressures of benzene and methylbenzene are 1521 and 570 mmHg respectively at the same temperature.
Assuming the liquid mixture obeys Raoult's law, calculate

(i) vapour pressure of the liquid mixture. (03 marks)

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(ii) the mole fraction of each component in the vapour phase.(02 mark)

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3. State what would be observed and write equation(s) for the reaction(s) that would take place when to a solution of manganese(II) sulphate was added:

(a) dilute sodium hydroxide solution drop wise until in excess.

Observation (1½ marks)

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Equation(s) (2½ marks)

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(b) concentrated nitric acid followed by lead(IV) oxide and the mixture warmed.

Observation (0½ marks)

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Equation(s) (1½ marks)

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4. (a) Derive an expression for the half life of a first order reaction:

$2.303 \log\left(\frac{a_o}{a_t}\right) = \lambda t$; where a_o is the initial concentration of the reactant and a_t is the concentration at time, t. (02 marks)

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(b) The rate constant of a first order reaction is $3.466 \times 10^{-3} s^{-1}$.

(i) Calculate the half life of the reaction. (01 mark)

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(ii) Determine the taken for the reaction to be 60% complete. (1½ marks)

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5. Name a reagent that can be used to differentiate be the following pairs of ions. State what would be observed if each ion is treated with the reagent you have named.

(a) $MnO_4^-(aq)$ and $MnO_4^{2-}(aq)$

Reagent (03 marks)

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Observation

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(b) $HCOO^-(aq)$ and $CH_3COO^-(aq)$

Reagent (03 marks)

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Observation

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6. 10 cm^3 of a hydrocarbon **Z** (C_xH_y) was exploded with 33 cm^3 of oxygen. On cooling to room temperature, the residual gases occupied 28 cm^3. When the residual gases were passed through potassium hydroxide solution, the volume reduced to 8 cm^3.

(a) Determine the molecular formula of **Z**. (03 mark)

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(b) Write the name and structural formula of **Z**. (01 mark)

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(c) Compound **Z** was bubbled into a solution mixture of silver nitrate and excess dilute ammonia.

(i) State what was observed (01 mark)

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(ii) Write equation for the reaction that took place. (01 mark)

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7. Write equation(s) to show how iodoethane can be converted to

(a) ethene. (1½ marks)

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(b) methylethanoate. (2½ marks)

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(c) butane. (01 mark)

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8. (a) Write

(i) equation for hydrolysis of sodium benzoate. (1½ marks)

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(ii) the expression for the hydrolysis constant of sodium benzoate (01 mark)

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(c) The pH of a 0.2 M aqueous solution of sodium benzoate at $25^{o}C$ is 8.75. Calculate the

(i) degree of hydrolysis of the salt. (02 mark)

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(ii) hydrolysis constant, K_h of the salt.

(1½ marks)

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9. Write equation for the reaction between

(a) iron(III) chloride solution and sodium hydrogencarbonate solution.

(1½ mark)

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(b) acidified potassium dichromate(VII) and hydrogen peroxide. (1½ marks)

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(c) 2,4-dinitrophenylhydrazine and ethanal. (1½ marks)

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Section B: (54 marks)

Answer **six** questions from this section.

Additional questions will **not** be marked.

10. Lithium is in Group I of the Periodic Table but it shows some similarities with magnesium which is in group II of the Periodic Table.

(a) Give **two** reasons why magnesium shows similarities with lithium.

(02 marks)

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(b) State **three** properties in which lithium and magnesium show similarities. (03 marks)

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(c) Write equation for the reaction between

(i) magnesium and dilute nitric acid. (01 mark)

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(ii) magnesium and steam. (01 mark)

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(iii) beryllium and sodium hydroxide. (02 marks)

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11. Write equation and mechanism for the reaction between

(a) a hot mixture of benzoic acid and methanol in the presence of concentrated sulphuric acid. (04 marks)

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(b) propanone and a saturated solution of sodium hydrogensulphite. (2½ marks)

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(c) hot mixture of phenol and bromoethane in presence of dilute sodium hydroxide solution. (2½ marks)

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12.(a) Equations for some half cell reactions are given below:

	E^{θ}/V
$H_2O_2(aq) + 2H^+(aq) + 2e^- \rightarrow 2H_2O(l)$	$+1.77$
$Sn^{4+}(aq) + 2e^- \rightarrow Sn^{2+}(aq)$	$+0.15$

(i) Write the cell notation for the cell formed by combining the two half reactions. (01 mark)

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(ii) Write equation for the overall cell reaction. (01 mark)

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(iii) Calculate the e.m.f of the cell. (1½ marks)

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(iv) State whether the cell reaction is feasible or not. Give a reason for your answer. (01 mark)

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(b) During the extraction of aluminum, a current of 0.2 amperes was passed through aluminum sulphate solution for one hour.

(i) Write equation for the reaction that took place at the cathode. (01 mark)

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(ii) Calculate the mass of aluminium produced. (02 marks)

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(c) State **one**

(i) application of electrode potentials. (0½ mark)

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(ii) way in which an electrolytic cell differs from an e.m.f cell. (01 mark)

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13. Carbon, silicon, germanium, tin and lead are in group IV of the Periodic Table.

(a) State

(i) the common oxidation states exhibited by the elements in their ions or compounds. (01 mark)

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(i) how the stability of the oxidation states of the elements in (a)(i) vary down the group. (01 mark)

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(b) Give a reason for your answer in (a)(ii). (01 mark)

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(c) Compare the thermal stabilities of carbon tetrachloride and lead tetrachloride. (Include equation(s) of reaction(s) if any) (2½ marks)

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(d) Describe the reaction if any between each of carbon tetrachloride and lead tetrachloride with water.

(i) Carbon tetrachloride. (01 mark)

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(ii) Lead tetrachloride. (2½ marks)

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14. Compound **R**, C_4H_6O forms a yellow precipitate with Brady's reagent but does not react with Tollen's reagent and turns reddish brown bromine water to colourless.

(a) Identify **R**. (01 mark)

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(b) Write equation for the reaction between **R** and

(i) Brady's reagent. (01 mark)

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(ii) bromine water. (01 mark)

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(c) Write a mechanism for the reaction that took place in (b)(ii). (02 marks)

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(d) Write equation to show how **R** can be converted into ethanol. (04 marks)

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15. (a) Draw the structure and name the shape of each of species in the table below. (4½ marks)

Species	**Structure**	**Shape**
PCl_5		
NO_3^-		
NH_4^+		

(b) Write equation for reaction between

(i) phosphorus pentachloride and water. (1½ marks)

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(ii) concentrated nitric acid and copper. (1½ marks)

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(iii) ammonium chloride solution and solid sodium sulphite. (1½ marks)

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16.(a) Define the following terms

(i) **Order of a reaction**. (01 mark)

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(ii) **Half life of a reaction**. (01 mark)

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(b) The table shows the kinetic data for the following reaction.

$$2H_2O_2(aq) \xrightarrow{suitable\ conditions} 2H_2O(l) + O_2(g)$$

$[H_2O_2]/mol\ dm^{-3}$	0.100	0.175	0.250	0.300
Initial rate $\times 10^{-4}/mol\ dm^{-3}s^{-1}$	0.593	1.040	1.480	1.820

(i) Plot a graph of initial rate against concentration in the space provided. (03 marks)

(ii) Determine the order of the reaction with respect to hydrogen peroxide. (01 mark)

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(iii) Determine the rate constant for the reaction and state its units. (01 mark)

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(iv) Write equation for the rate of the reaction in terms of concentration of hydrogen peroxide. (01 mark)

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17. (a) State what is meant by the term **soap**. (01 mark)

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(b) Describe briefly how soap can be prepared from named oil. (02 marks)

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(c) Write

(i) the general equation for the reaction leading to the formation of soap. (01 mark)

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(ii) the equation for reaction between an aqueous soap solution and aluminium powder. (1½ marks)

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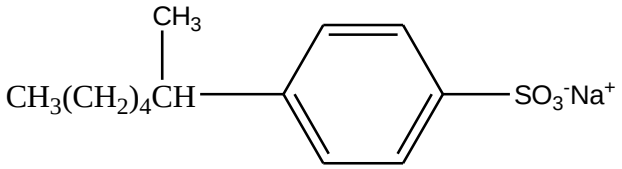
(c) A soapless detergent has the following structural formula.



(i) Write equation to show how the detergent can be synthesized from $CH_3(CH_2)_4CH = CH_2$ and benzene as the starting materials.

(02 marks)

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(ii) Name **one** substance that is added to the detergents to improve on their action during washing. (0½ mark)

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(iii) State how the named substance in (c)(ii) improves the action of the detergents during washing. (01 mark)

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End